

Geometric progression

Consider the sequence $\{a_k\}_{k=0}^n = \{a \cdot r^k\}_{k=0}^n$, where $a, r \in \mathbb{R}$. Find $\sum_{k=0}^n a_k$.

$$\{a_0, a_1, a_2, \dots, a_n\}$$

$$\{ar^0, ar^1, ar^2, \dots, ar^n\}$$

$$\{a, ar, ar^2, \dots, ar^n\}$$

$$\sum_{k=0}^n a_k = a + ar + ar^2 + \dots + ar^n$$
$$= a(1 + r + r^2 + \dots + r^n)$$

$$\boxed{r=1} \rightarrow \sum_{k=0}^n a_k = a(1 + \overbrace{1 + \dots + 1}^n) = a(1+n)$$

$$\sum_{k=0}^n a_k = a + \underbrace{ar + ar^2 + \dots + ar^n}_{=}$$

$$\boxed{r \neq 1}$$

$$r \sum_{k=0}^n a_k = \underbrace{ar + ar^2 + ar^3 + \dots + ar^n}_{\left[\sum_{k=0}^n a_k \right] - a} + ar^{n+1}$$

$$r \sum_{k=0}^n a_k = \sum_{k=0}^n a_k - a + ar^{n+1}$$

$$r \sum_{k=0}^n a_k - \sum_{k=0}^n a_k = ar^{n+1} - a$$

$$(r-1) \sum_{k=0}^n a_k = a(r^{n+1} - 1)$$

$$\sum_{k=0}^n a_k = \frac{a(r^{n+1} - 1)}{(r-1)}$$

Geometric progression

We proved that for $a, r \in \mathbb{R}$:

$$\sum_{k=0}^n ar^k = \begin{cases} \frac{a(r^{n+1} - 1)}{r - 1} & \text{if } r \neq 1 \\ (n + 1)a & \text{if } r = 1 \end{cases}$$

How do we compute $\sum_{k=0}^{\infty} a_k$?

Infinite sums (example)

Given $x \in \mathbb{R}$ such that $|x| < 1$, compute $\sum_{j=0}^{\infty} \underline{x^j}$.

$$a=1$$

$$r=x$$

$$\sum_{j=0}^n r^j = \frac{\overset{\text{small}}{\underbrace{r^{n+1}}_{\text{large \#}}}}{r-1}$$

if

$\underbrace{r^{n+1}}_{\text{large \#}}, |r| \leq 1$
converges to 0

$$\lim_{n \rightarrow \infty} \sum_{j=0}^n r^j = \frac{-1}{r-1} = \frac{1}{1-r}$$

Some summation formulas

Sum	Closed Form
$\sum_{k=0}^n ar^k (r \neq 0, 1)$	$\frac{ar^{n+1} - a}{r - 1}$
$\sum_{k=1}^n k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^n k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^n k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, x < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty} kx^{k-1}, x < 1$	$\frac{1}{(1-x)^2}$

Telescopic sum

Given a sequence $\{a_j\}$, let's compute $\sum_{j=1}^n (a_j - a_{j-1})$.

$$\begin{aligned} &= (\cancel{a_1} - a_0) + (a_1 - \cancel{a_2}) + (\cancel{a_2} - a_3) + \dots + (a_n - \cancel{a_{n-1}}) \\ &= a_n - a_0 \end{aligned}$$

$$\frac{1}{k} - \frac{1}{k+1} = \frac{k+1-k}{k(k+1)} = \frac{1}{k(k+1)}$$

Example

Compute $\sum_{k=1}^n \frac{1}{k(k+1)} = \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right)$

$$\begin{aligned} &= \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ &= 1 - \frac{1}{n+1} \end{aligned}$$